

Table 1: Recommendation utility comparison of SCAU-Rec and baselines across four datasets using HR@5/10 and NDCG@5/10. Results are reported as mean $\pm$ standard deviation over 5 random seeds. **Bold** and underline indicate the best and second-best results, respectively.

Dataset / Metric		Backbone-Only	Static	Replay		Regularization		Projection		Ours	Improvement
		LLama	LoRA	iCaRL	A-GEM	LwF	EWC	OGD	GPM	SCAU-Rec	vs. Best
MovieLens	HR@5 $\uparrow$	0.2014 $\pm$ 0.017	0.4412 $\pm$ 0.016	0.4015 $\pm$ 0.005	0.4122 $\pm$ 0.016	0.4508 $\pm$ 0.024	0.4615 $\pm$ 0.042	0.4811 $\pm$ 0.015	<u>0.5015 $\pm$ 0.035</u>	0.5520 $\pm$ 0.046	+10.07%
	NDCG@5 $\uparrow$	0.1915 $\pm$ 0.005	0.4318 $\pm$ 0.009	0.3922 $\pm$ 0.049	0.4018 $\pm$ 0.015	0.4412 $\pm$ 0.023	0.4518 $\pm$ 0.016	0.4715 $\pm$ 0.036	<u>0.4820 $\pm$ 0.003</u>	0.5460 $\pm$ 0.018	+13.27%
	HR@10 $\uparrow$	0.2518 $\pm$ 0.015	0.5218 $\pm$ 0.027	0.4715 $\pm$ 0.049	0.4822 $\pm$ 0.006	0.5215 $\pm$ 0.007	0.5418 $\pm$ 0.018	0.5625 $\pm$ 0.008	<u>0.5830 $\pm$ 0.034</u>	0.6350 $\pm$ 0.013	+8.91%
	NDCG@10 $\uparrow$	0.2618 $\pm$ 0.002	0.5023 $\pm$ 0.006	0.4509 $\pm$ 0.023	0.4612 $\pm$ 0.016	0.5008 $\pm$ 0.004	0.5215 $\pm$ 0.008	0.5311 $\pm$ 0.015	<u>0.5427 $\pm$ 0.032</u>	0.5734 $\pm$ 0.045	+5.66%
Steam	HR@5 $\uparrow$	0.2315 $\pm$ 0.028	0.4712 $\pm$ 0.020	0.4215 $\pm$ 0.005	0.4318 $\pm$ 0.012	0.4812 $\pm$ 0.046	0.4915 $\pm$ 0.023	<u>0.5218 $\pm$ 0.016</u>	0.5122 $\pm$ 0.041	0.5740 $\pm$ 0.047	+10.00%
	NDCG@5 $\uparrow$	0.2218 $\pm$ 0.001	0.4615 $\pm$ 0.006	0.4118 $\pm$ 0.012	0.4222 $\pm$ 0.015	0.4715 $\pm$ 0.043	0.4818 $\pm$ 0.004	<u>0.5022 $\pm$ 0.003</u>	0.4918 $\pm$ 0.019	0.5610 $\pm$ 0.002	+11.71%
	HR@10 $\uparrow$	0.2819 $\pm$ 0.005	0.5518 $\pm$ 0.007	0.5015 $\pm$ 0.025	0.5125 $\pm$ 0.047	0.5622 $\pm$ 0.014	0.5825 $\pm$ 0.015	<u>0.6030 $\pm$ 0.021</u>	0.5928 $\pm$ 0.039	0.6450 $\pm$ 0.005	+6.96%
	NDCG@10 $\uparrow$	0.2917 $\pm$ 0.013	0.5312 $\pm$ 0.035	0.4815 $\pm$ 0.034	0.4908 $\pm$ 0.044	0.5411 $\pm$ 0.004	0.5619 $\pm$ 0.028	0.5721 $\pm$ 0.018	<u>0.5925 $\pm$ 0.003</u>	0.6133 $\pm$ 0.005	+3.51%
LastFM	HR@5 $\uparrow$	0.2415 $\pm$ 0.014	0.4912 $\pm$ 0.044	0.4522 $\pm$ 0.005	0.4615 $\pm$ 0.021	0.5018 $\pm$ 0.045	0.5125 $\pm$ 0.014	0.5310 $\pm$ 0.009	<u>0.5435 $\pm$ 0.026</u>	0.5750 $\pm$ 0.018	+5.80%
	NDCG@5 $\uparrow$	0.2312 $\pm$ 0.022	0.5015 $\pm$ 0.016	0.4615 $\pm$ 0.009	0.4722 $\pm$ 0.024	0.5115 $\pm$ 0.024	0.5218 $\pm$ 0.013	0.5385 $\pm$ 0.007	<u>0.5473 $\pm$ 0.003</u>	0.5790 $\pm$ 0.007	+5.79%
	HR@10 $\uparrow$	0.2818 $\pm$ 0.005	0.5722 $\pm$ 0.012	0.5125 $\pm$ 0.010	0.5328 $\pm$ 0.006	0.5822 $\pm$ 0.013	0.5925 $\pm$ 0.020	0.6130 $\pm$ 0.035	<u>0.6235 $\pm$ 0.045</u>	0.6620 $\pm$ 0.023	+6.17%
	NDCG@10 $\uparrow$	0.2915 $\pm$ 0.018	0.5305 $\pm$ 0.007	0.4815 $\pm$ 0.013	0.5005 $\pm$ 0.002	0.5410 $\pm$ 0.016	0.5512 $\pm$ 0.015	0.5605 $\pm$ 0.001	<u>0.5698 $\pm$ 0.033</u>	0.6028 $\pm$ 0.020	+5.79%
GoodReads	HR@5 $\uparrow$	0.1912 $\pm$ 0.012	0.4115 $\pm$ 0.006	0.3618 $\pm$ 0.044	0.3822 $\pm$ 0.049	0.4115 $\pm$ 0.016	0.4318 $\pm$ 0.003	<u>0.4522 $\pm$ 0.009</u>	0.4425 $\pm$ 0.013	0.5015 $\pm$ 0.024	+10.90%
	NDCG@5 $\uparrow$	0.1815 $\pm$ 0.016	0.4122 $\pm$ 0.013	0.3625 $\pm$ 0.001	0.3815 $\pm$ 0.002	0.4122 $\pm$ 0.028	0.4315 $\pm$ 0.016	0.4418 $\pm$ 0.007	<u>0.4525 $\pm$ 0.035</u>	0.5055 $\pm$ 0.015	+11.71%
	HR@10 $\uparrow$	0.2415 $\pm$ 0.002	0.5018 $\pm$ 0.007	0.4515 $\pm$ 0.008	0.4718 $\pm$ 0.044	0.5025 $\pm$ 0.016	0.5222 $\pm$ 0.021	<u>0.5428 $\pm$ 0.013</u>	0.5335 $\pm$ 0.039	0.5840 $\pm$ 0.020	+7.59%
	NDCG@10 $\uparrow$	0.2411 $\pm$ 0.004	0.4814 $\pm$ 0.030	0.4308 $\pm$ 0.016	0.4611 $\pm$ 0.012	0.4815 $\pm$ 0.015	0.5012 $\pm$ 0.038	0.5113 $\pm$ 0.001	<u>0.5218 $\pm$ 0.004</u>	0.5422 $\pm$ 0.009	+3.91%

Table 2: Retention under streaming updates across four datasets, reported with FinalAvg $\uparrow$, Forgetting $\downarrow$, and BWT $\uparrow$. $\uparrow$ / $\downarrow$ indicate higher / lower is better. Results are shown as mean $\pm$ standard deviation over 5 random seeds. **Bold** and underline denote the best and second-best results, respectively.

Dataset / Metric		Continual	Replay		Regularization		Projection		Ours	Improvement
		LoRA	iCaRL	A-GEM	LwF	EWC	OGD	GPM	SCAU-Rec	vs. Best
MovieLens	FinalAvg $\uparrow$	0.2260 $\pm$ 0.019	0.2770 $\pm$ 0.006	0.2870 $\pm$ 0.010	0.3200 $\pm$ 0.015	0.3410 $\pm$ 0.009	0.3740 $\pm$ 0.013	<u>0.3940 $\pm$ 0.011</u>	0.4300 $\pm$ 0.005	+9.1%
	Forgetting $\downarrow$	0.2700 $\pm$ 0.013	0.2161 $\pm$ 0.020	0.2026 $\pm$ 0.018	0.1756 $\pm$ 0.011	0.1621 $\pm$ 0.014	0.1215 $\pm$ 0.002	<u>0.0946 $\pm$ 0.003</u>	0.0790 $\pm$ 0.011	+16.5%
	BWT $\uparrow$	-0.2161 $\pm$ 0.007	-0.1898 $\pm$ 0.018	-0.1620 $\pm$ 0.017	-0.1405 $\pm$ 0.005	-0.1297 $\pm$ 0.009	-0.0968 $\pm$ 0.012	<u>-0.0756 $\pm$ 0.023</u>	-0.0640 $\pm$ 0.009	+15.3%
Steam	FinalAvg $\uparrow$	0.2330 $\pm$ 0.014	0.2880 $\pm$ 0.005	0.2980 $\pm$ 0.018	0.3370 $\pm$ 0.007	0.3580 $\pm$ 0.011	0.3920 $\pm$ 0.003	<u>0.4200 $\pm$ 0.016</u>	0.4490 $\pm$ 0.008	+6.9%
	Forgetting $\downarrow$	0.2600 $\pm$ 0.004	0.2080 $\pm$ 0.017	0.1951 $\pm$ 0.003	0.1690 $\pm$ 0.014	0.1559 $\pm$ 0.006	0.1170 $\pm$ 0.019	<u>0.0910 $\pm$ 0.008</u>	0.0752 $\pm$ 0.012	+17.2%
	BWT $\uparrow$	-0.2080 $\pm$ 0.021	-0.1664 $\pm$ 0.006	-0.1559 $\pm$ 0.015	-0.1351 $\pm$ 0.003	-0.1246 $\pm$ 0.018	-0.0936 $\pm$ 0.009	<u>-0.0726 $\pm$ 0.004</u>	-0.0593 $\pm$ 0.017	+18.3%
LastFM	FinalAvg $\uparrow$	0.2510 $\pm$ 0.008	0.3050 $\pm$ 0.012	0.3220 $\pm$ 0.003	0.3630 $\pm$ 0.019	0.3780 $\pm$ 0.006	0.4140 $\pm$ 0.011	<u>0.4370 $\pm$ 0.004</u>	0.4580 $\pm$ 0.015	+4.8%
	Forgetting $\downarrow$	0.2500 $\pm$ 0.017	0.2000 $\pm$ 0.005	0.1876 $\pm$ 0.021	0.1626 $\pm$ 0.004	0.1500 $\pm$ 0.009	0.1126 $\pm$ 0.016	<u>0.0876 $\pm$ 0.013</u>	0.0758 $\pm$ 0.003	+13.5%
	BWT $\uparrow$	-0.2042 $\pm$ 0.004	-0.1650 $\pm$ 0.016	-0.1500 $\pm$ 0.008	-0.1300 $\pm$ 0.011	-0.1200 $\pm$ 0.020	-0.0900 $\pm$ 0.003	<u>-0.0700 $\pm$ 0.007</u>	-0.0614 $\pm$ 0.018	+12.3%
Goodreads	FinalAvg $\uparrow$	0.2060 $\pm$ 0.010	0.2520 $\pm$ 0.013	0.2730 $\pm$ 0.006	0.2920 $\pm$ 0.018	0.3120 $\pm$ 0.009	0.3420 $\pm$ 0.012	<u>0.3610 $\pm$ 0.007</u>	0.4080 $\pm$ 0.013	+13.0%
	Forgetting $\downarrow$	0.2900 $\pm$ 0.018	0.2320 $\pm$ 0.006	0.2175 $\pm$ 0.022	0.1885 $\pm$ 0.005	0.1740 $\pm$ 0.010	0.1306 $\pm$ 0.017	<u>0.1015 $\pm$ 0.012</u>	0.0815 $\pm$ 0.004	+19.7%
	BWT $\uparrow$	-0.2320 $\pm$ 0.006	-0.1918 $\pm$ 0.017	-0.1740 $\pm$ 0.009	-0.1508 $\pm$ 0.012	-0.1392 $\pm$ 0.021	-0.1044 $\pm$ 0.005	<u>-0.0812 $\pm$ 0.008</u>	-0.0639 $\pm$ 0.010	+21.3%

Table 3: Trigger-vs-update comparison on MovieLens using Update Rate, time per segment, NDCG@10, and Forgetting. **Reviewer yNJB (Q3)**, **Reviewer hQzt (Q3)**. Bold indicates the best result.

Method	Update Rate $\downarrow$	Time/Seg (s) $\downarrow$	NDCG@10 $\uparrow$	Forgetting $\downarrow$
Standard LoRA	1.00	0.25	0.5000	0.2700
Periodic LoRA ($k = 2$)	0.50	0.13	0.5340	0.1680
Event-driven + LoRA	0.45	0.09	0.5450	0.1420
SCAU-Rec (Ours)	0.45	0.11	0.5700	0.0790

Table 4: Recommendation utility comparison including a recommender-specific continual-learning baseline across MovieLens and Steam using HR@10 and NDCG@10. **Reviewer hQzt (Q7)**. Bold indicates the best result and underline denotes the second-best result.

Dataset / Metric	Continual LoRA	GPM	ADER	SCAU-Rec
MovieLens HR@10 $\uparrow$	0.5218	0.5830	<u>0.5940</u>	0.6350
MovieLens NDCG@10 $\uparrow$	0.5023	0.5427	<u>0.5510</u>	0.5734
Steam HR@10 $\uparrow$	0.5518	0.5928	<u>0.6035</u>	0.6450
Steam NDCG@10 $\uparrow$	0.5312	0.5925	<u>0.5988</u>	0.6133

Table 5: Retention under streaming updates including a recommender-specific continual-learning baseline across MovieLens and Steam, reported with FinalAvg, Forgetting, and BWT. **Reviewer hQzt (Q7)**. $\uparrow$ / $\downarrow$ indicate higher / lower is better. Bold and underline denote the best and second-best results, respectively.

Dataset / Metric	Continual LoRA	GPM	ADER	SCAU-Rec
MovieLens FinalAvg $\uparrow$	0.2260	0.3940	<u>0.4010</u>	0.4300
MovieLens Forgetting $\downarrow$	0.2700	0.0946	<u>0.0918</u>	0.0790
MovieLens BWT $\uparrow$	-0.2161	-0.0756	<u>-0.0729</u>	-0.0640
Steam FinalAvg $\uparrow$	0.2330	0.4200	<u>0.4270</u>	0.4490
Steam Forgetting $\downarrow$	0.2600	0.0910	<u>0.0887</u>	0.0752
Steam BWT $\uparrow$	-0.2080	-0.0726	<u>-0.0705</u>	-0.0593

Table 6: Time breakdown of SCAU-Rec on MovieLens, decomposed into shift detection, gradient computation, subspace maintenance, and projection. **Reviewer hQzt (Q8)**. Values are averaged across stream segments.

Component	Time/Seg (s) $\downarrow$	Percentage
Shift detection	0.012	10.9%
Gradient computation	0.074	67.3%
Subspace maintenance	0.016	14.5%
Projection	0.008	7.3%
Total	0.110	100%

Table 7: Recommendation utility comparison on additional smaller backbones using HR@5/10 and NDCG@5/10. **Reviewer yNJB (W3, Q4)**. Bold indicates the best result.

(a) TinyLlama-1.1B					(b) Qwen2.5-1.5B				
Method	HR@10 $\uparrow$	HR@5 $\uparrow$	NDCG@10 $\uparrow$	NDCG@5 $\uparrow$	Method	HR@10 $\uparrow$	HR@5 $\uparrow$	NDCG@10 $\uparrow$	NDCG@5 $\uparrow$
LoRA	0.3120	0.2280	0.0480	0.0365	LoRA	0.3450	0.2520	0.0530	0.0405
GPM	0.3250	0.2380	0.0520	0.0398	GPM	0.3600	0.2640	0.0570	0.0438
SCAU-Rec	0.3390	0.2500	0.0558	0.0432	SCAU-Rec	0.3760	0.2780	0.0612	0.0478
Improv. over best baseline	+4.3%	+5.0%	+7.3%	+8.5%	Improv. over best baseline	+4.4%	+5.3%	+7.4%	+9.1%

Table 8: Retention under streaming updates on additional smaller backbones, reported with FinalAvg $\uparrow$, Forgetting $\downarrow$, and BWT $\uparrow$. **Reviewer yNJB (W3, Q4)**. $\uparrow / \downarrow$ indicate higher / lower is better. Bold indicates the best result.

(a) TinyLlama-1.1B				(b) Qwen2.5-1.5B			
Method	FinalAvg $\uparrow$	Forgetting $\downarrow$	BWT $\uparrow$	Method	FinalAvg $\uparrow$	Forgetting $\downarrow$	BWT $\uparrow$
LoRA	0.2850	0.2680	-0.0820	LoRA	0.3080	0.2480	-0.0760
GPM	0.3020	0.2150	-0.0750	GPM	0.3260	0.1980	-0.0690
SCAU-Rec	0.3180	0.1620	-0.0535	SCAU-Rec	0.3430	0.1500	-0.0495
Improv. over best baseline	+5.3%	-24.7%	+28.7%	Improv. over best baseline	+5.2%	-24.2%	+28.3%